Lecture plan · for the lecturer

Science stand-ins, and one machine

University, scale 4

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Fast where it was tested
- Not the experiment

Activity: In the lab

1. Wide layer, both splits
2. Large batch, narrow layer: split the examples

Discussion

- What does a bigger machine not answer?

Exit ticket

- Wide layer: which split fits?

Try it in your next lecture
Refuse the bigger-GPU sentence in the first five minutes.

Seminar sheet · for students

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University, scale 4

Name _____ Group _____ Date _____

1 **A layer does not fit on one machine because it is wide. You…**

◯ Split the layer ◯ Use a bigger batch ◯ Only split the examples

2 **A fast stand-in for a lab measurement is the experiment.**

◯ True ◯ False

3 **Match the split.**

Data parallel	• • • •	Each machine holds part of the layer
Model parallel	• • • •	One machine, so this lab is a diagram
One phone	• • • •	Each machine holds the whole model and some examples

4 **Order a decision about a wide layer.**

☐ Split the layer if that still does not fit

☐ Write the memory of one step

☐ Try splitting the examples

Answers · for the lecturer

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1 **Split the layer**

Splitting the examples still copies the whole layer onto each machine, and a bigger batch only adds activations.

2 **False**

It is useful where it was tested. It is not the measurement.

3 **Data parallel — Each machine holds the whole model and some examples**
Model parallel — Each machine holds part of the layer
One phone — One machine, so this lab is a diagram

Memory of a step is the weights, their gradients and the optimizer's state, plus the activations; the activations grow with the batch.

4 **Top to bottom: 3, 1, 2**

Measure first, before you name a bigger machine.

For you to think about

What formula must be on the board first?